

A compact semialgebraic function without exponential Fourier decay

Disproof of Conjecture 00000008046

gaochengzhecpu

Abstract

The characteristic function of $[0, 1]$ is definable, compactly supported, and subanalytic. Nevertheless, its Fourier transform has modulus $1/(\pi(n + 1/2))$ at frequency $n + 1/2$. This excludes every eventual bound $Ce^{-c|\xi|}$ with $c > 0$, disproving the claimed implication from subanalyticity to exponential Fourier decay.

The function satisfies the stated hypotheses

The source concerns definable compactly supported functions and claims that exponential Fourier decay holds if and only if the function is subanalytic. No continuity or differentiability hypothesis is imposed. Consider

$$f(x) = \mathbf{1}_{[0,1]}(x).$$

Its support is exactly the compact interval $[0, 1]$. Its real graph is

$$\{(x, y) : 0 \leq x \leq 1, y = 1\} \cup \{(x, y) : (x < 0 \text{ or } x > 1), y = 0\}.$$

This finite Boolean combination of polynomial conditions makes the graph semialgebraic, hence definable in the ordered real field and in the expansions appearing in the source. Polynomial functions are real analytic, so the displayed description is also semianalytic. Locally a semianalytic set is subanalytic by the definition using projections of relatively compact semianalytic sets (restrict to a bounded coordinate neighborhood and use the identity projection). Thus the graph, and hence the function in the standard graph sense, is subanalytic. The discontinuities at $0, 1$ do not violate any hypothesis in the source. Also $f \in L^1(\mathbb{R})$ since $\int |f| = 1$.

The genuine Fourier integral

Use the convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx.$$

For $\xi \neq 0$, direct integration on the support gives

$$\widehat{f}(\xi) = \int_0^1 e^{-2\pi i x \xi} dx = \frac{e^{-2\pi i \xi} - 1}{-2\pi i \xi}.$$

At $\xi_n = n + 1/2$ for $n = 0, 1, \dots$, the exponential is -1 . Consequently

$$\widehat{f}(\xi_n) = \frac{-i}{\pi \xi_n}, \quad |\widehat{f}(\xi_n)| = \frac{1}{\pi \xi_n}.$$

These are exact integral values, not an approximation or an assumption about a generic Fourier decay model.

Failure of every exponential bound

Suppose constants $C > 0$, $c > 0$, and $R \in \mathbb{R}$ gave $|\widehat{f}(\xi)| \leq Ce^{-c|\xi|}$ for every $|\xi| \geq R$. For sufficiently large n , the positive frequency ξ_n exceeds R , so the exact values above imply

$$1 \leq \pi C \xi_n e^{-c\xi_n}.$$

But $xe^{-cx} \rightarrow 0$ as $x \rightarrow +\infty$ for every $c > 0$. The right side therefore tends to zero, a contradiction. Indeed the same contradiction excludes a bound on just the positive tail $\xi \geq R$. The alternative Fourier convention $\int f(x)e^{-ix\xi} dx$ only rescales frequency and gives the same failure. Thus this admissible subanalytic function disproves the stated “if” direction, which suffices to refute the equivalence.

Correspondence with Lean

The function `f` in `lean/Main.lean` is the actual interval indicator, viewed in $\mathbb{C}$. The theorem `graph_description` proves the displayed finite polynomial description, including that its imaginary coordinate is zero. This is the explicit certificate for the semialgebraic and subanalytic hypotheses. Those general geometric categories are not renamed or replaced by an unrelated predicate in the Lean file.

The theorem `compact_support` proves genuine Mathlib compact support. The theorem `integral_formula` evaluates `Real.fourierIntegral` using its actual Bochner integral, restriction to $[0, 1]$, the equality of closed/open-endpoint integrals under Lebesgue measure, and the exponential antiderivative. The theorems `fourier_at_frequency` and `norm_fourier_at_frequency` give the exact half-integer values. Finally `no_exponential_bound` quantifies over arbitrary real C, R and every $c > 0$, using the actual exponential limit and arbitrarily large half-integer frequencies. It is stronger than the contradiction needed for positive C .

Reproduction

The project pins Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build` and `lake env lean -DwarningAsError=true Main.lean`. On a new machine, `lake exe cache get` retrieves the official pinned dependency artifacts. The submission itself is checked in a fresh build directory. Printed dependencies use only the standard `propext`, `Classical.choice`, and `Quot.sound`. No additional axioms, proof placeholders, or numerical oracles are used. Run `tectonic main.tex` to reproduce the PDF; no auxiliary numerical script is needed. Build logs, source hashes, and visual-review records are included under `verification/`.

Conjecture source

The Last Math Competition, Conjecture 00000008046. The exact bilingual source and its commit and SHA-256 are preserved in this submission.